

Hill Climbing Techniques

20MCA188 Artificial Intelligence

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Simple Hill Climbing

1. Evaluate the initial state. If it is also a goal state, then return it and quit.
2. Loop until a solution is found or until there are no new operators left to be applied in the current state:
 - 2.1 Select an operator that has not yet been applied to the current state and apply it to produce a new state.
 - 2.2 Evaluate the new state.
 - 2.2.1 If it is a goal state, return it and quit.
 - 2.2.2 If it is better than the current state, then make it the current state.
 - 2.2.3 If it is not better than the current state, continue in the loop.

Steepest-Ascent Hill Climbing

1. Evaluate the initial state. If it is also a goal state, then return it and quit. Otherwise, continue with the initial state as the current state.
2. Loop until a solution is found or until a complete iteration produces no change to the current state:
 - 2.1 Let SUCC be a state such that any possible successor of the current state will be better than SUCC.
 - 2.2 For each operator that applies to the current state do:
 - 2.2.1 Apply the operator and generate a new state.
 - 2.2.2 Evaluate the new state. If it is a goal state, then return it and quit. If it is better than SUCC, then set SUCC to this state.
 - 2.3 If SUCC is better than the current state, then set current state to SUCC.
 - 2.4 If there is no state better than the current state, then return the current state and quit.

Designing Heuristics-8-Puzzle Problem

- 8 numbered tiles and one blank space.
- Goal: Rearrange the tiles into the correct order.
- Various heuristics help estimate the cost to the goal.

2	8	3
1	6	4
7		5

Heuristic 1: Misplaced Tiles (h_1)

Definition: Counts the number of tiles in incorrect positions.

$$h_1 = \text{number of misplaced tiles}$$

Advantages: Simple and fast to compute.

Disadvantages: Ignores how far tiles are from their target.

Example:

1 2 3

5 4 7

6 8 _

If goal is:

1 2 3

4 5 6

7 8 _

$$h_1 = 4$$

Heuristic 2: Manhattan Distance (h_2)

Definition: Sum of horizontal and vertical moves needed to reach the goal position.

$$h_2 = \sum_{\text{each tile}} \text{distance to goal}$$

Advantages: More accurate than misplaced tiles.

Disadvantages: Slightly slower than h_1 , but still efficient.

Example: Let tile at (1,1) needs to go to (3,3):

$$|1 - 3| + |1 - 3| = 4$$

Heuristic h_3 for 8-Puzzle

- In sliding tile puzzles, tiles cannot jump over each other.
- h_2 (e.g., Manhattan Distance) underestimates the real difficulty when two adjacent tiles need to swap places.
- h_3 adds a penalty for these "adjacent tile conflicts."

$$h_3(\text{node}) = h_2(\text{node}) + 2 \cdot k(\text{node})$$

Where:

- h_2 = Manhattan distance (basic heuristic)
- k = number of adjacent tile conflicts

Example Configuration

Goal State:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & - \end{bmatrix}$$

Current State:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 6 & 5 \\ 7 & 8 & - \end{bmatrix}$$

- Tile 5 and 6 are out of order.
- $h_2 = 2$ (Manhattan distance: Tile 5 needs 1 move, Tile 6 needs 1 move).
- $k = 1$ (one adjacent swap needed).

$$h_3 = h_2 + 2 \cdot k = 2 + 2 \cdot 1 = 4$$

Summary Table

Term	Meaning
h_2	Basic Manhattan Distance
k	Number of adjacent tile swaps needed
$2k$	Extra penalty for resolving conflicts
h_3	Total improved heuristic: $h_2 + 2k$

Comparison

Heuristic	Admissible	Computation	Accuracy
Misplaced Tiles	Yes	Very Fast	Low
Manhattan Distance	Yes	Fast	Good
Linear Conflict	Yes	Moderate	Better